

Phase equilibria of phenolic compounds in water or ethanol



Laëtitia Cesari, Sylvain Namysl, Laetitia Canabady-Rochelle, Fabrice Mutelet*

Université de Lorraine, Ecole Nationale Supérieure des Industries Chimiques, Laboratoire Réactions et Génie des Procédés (UMR CNRS 7274), 1 Rue Grandville, 54000 Nancy, France

ARTICLE INFO

Article history:

Received 9 June 2017

Received in revised form

3 September 2017

Accepted 11 September 2017

Available online 12 September 2017

Keywords:

Phase equilibria

Phenolic compounds

NRTL

Water

Ethanol

ABSTRACT

The mutual solubilities of water and eight phenolic compounds (*i.e.* phenol, guaiacol, syringol, pyrocatechol, *o*-, *m*-, *p*-cresol and vanillin) were investigated between 293.15 and 323.15K. The solubility of the phenolic compounds in water increases as follows: vanillin < syringol < guaiacol < *p*-cresol < *m*-cresol < *o*-cresol < phenol < pyrocatechol. For the molecules presenting liquid-liquid equilibria, the solubility of water into the phenolic phase increases as follows: guaiacol < *o*-cresol < *m*-cresol < *p*-cresol < syringol. The vapor-liquid equilibria of binary mixtures composed of ethanol and phenol or guaiacol or *o*-cresol were measured in a range of pressure from 0.09 bar to atmospheric pressure. Finally, the mixing enthalpies of the phenolic compounds with ethanol indicate that the reactions are exothermic. It was found that NRTL model is suitable for the representation of phase diagrams of systems containing phenolic compounds and water or ethanol.

© 2017 Elsevier B.V. All rights reserved.

1. Introduction

Most of the chemicals, such as phenolic compounds, are produced from the petrochemical industry. Yet, with the rise of global warming and the limited stocks of fossil fuels, there is an urgent need to find new ways of production for such compounds. For the last years, lignin has become an interesting tool for the production of monophenols [1]. Indeed, the thermochemical conversion processes of lignin can provide liquid fuels [2] as well as bio-oils rich in valuable chemicals. With specific bioactivities, such as antioxidant or antimicrobial properties, phenolic compounds are used as intermediate or raw material in the food, pharmaceutical and cosmetic industries [3–6].

Bio-oils can be produced by various thermochemical treatments, the famous ones being pyrolysis and liquefaction. Many studies have investigated the influence of solvent on the liquefaction process, including formic acid [7], water [8–10], methanol, 2-propanol [10], or ethanol [10–13]. In order to reduce the CO₂ emission and be more economically competitive, more and more processes are using solvents coming from biomass for liquefaction processes [14]. Besides, biomass pyrolysis also produces aromatic compounds-rich bio-oils. The condensed vapors obtained during a pyrolysis process are generally collected in solvents such as

aqueous solution or ethanol. The extraction of high value compounds from this diluted bio-oil is difficult to perform due to its thermal instability and its complex composition. Most of the processes generally start with the recovery of the solvent by distillation. Then, liquid-liquid extraction using water [15], basic aqueous solutions [16–18] or polar organic solvent [19] are predominantly preferred. Although the distillation of ethanol and the liquid-liquid extraction of phenolic compounds with water are regularly used in the extraction process, only few data are available in the literature [20–39]. The method and the range of temperature of the {phenolic compounds-water} systems found in the literature are presented in Table 1. No data has been found for the syringol-water system. The vapor-liquid equilibria (VLE) of the {phenol + ethanol} system were only investigated at atmospheric pressure [40,41], at 288.15K [42] and 293.15 K [43]. Only one isothermal VLE at 290.15 K can be found for the binary {guaiacol + ethanol} [38] and two sets of VLE data for the binary system {*o*-cresol + ethanol} (0.9576 bar [37] and 291.15K [39]).

To date, there is still a huge lack of information about the behavior of phenolic compounds with such solvents, especially for guaiacol and syringol. It is therefore difficult to determine the most adapted solvent for such extraction and to estimate its feasibility and extraction capacity.

In this work, we studied the behavior of eight phenolic compounds (*i.e.* phenol, guaiacol, syringol, pyrocatechol, *o*-cresol, *m*-cresol, *p*-cresol and vanillin) with water or ethanol. The mutual solubilities of the compounds with water were investigated as a

* Corresponding author.

E-mail address: fabrice.mutelet@univ-lorraine.fr (F. Mutelet).

Table 1

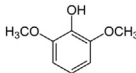
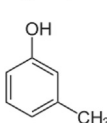
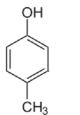
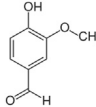
Temperature range and method of measurements for the {phenolic compounds–water} systems found in the literature.

System	Temperature	Method	Reference
Phenol–water	[280–345K]	Conductivity	[31,34]
Phenol–water	[274–338K]	Refractometry	[32]
Phenol–water	[293–339K]	Volumetry	[33]
Pyrocatechol–water	[314–377K]	Visual observation	[29]
Pyrocatechol–water	[295–326K]	/	[30]
Vanillin–water	[273–323K]	Visual observation	[26]
Vanillin–water	[273–303K]	Visual observation	[27]
Vanillin–water	[293–318K]	Shake flask method	[28]
Cresol–water	[273–450K]	Visual observation	[21]
Cresol–water	298K	Shake flask method	[22]
Cresol–water	293K	Turbidity	[23]
Cresol–water	[288; 298K]	/	[24]
m-cresol–water	[293–350K]	Chromatography	[25]
Cresol–water	[298–360K]	Turbidity	[36]
Guaiacol–water	298K	Shake flask method	[20]
Guaiacol–water	298K	Density measurements	[35]

function of temperature. Bubble curves of phenolic compounds with ethanol as well as their mixing enthalpy were also studied. The phase equilibrium data of the systems were used to regress parameters on the Non-Random Two-Liquids (NRTL) [44] thermodynamic models.

Table 2

Name, CAS number, abbreviation, molar mass, formula, source and purity of the used compounds.

Name and CAS	Abbreviation	Molar mass g.mol ⁻¹	Formula	Source and purity (wt%)
Phenol 108-95-2		94.11		Sigma Aldrich >99.5%
2-methoxyphenol 90-05-1	Guaiacol	124.14		Sigma Aldrich >99%
2,6-dimethoxyphenol 91-10-1	Syringol	154.16		Sigma Aldrich 99%
1,2-benzenediol 120-80-9	Pyrocatechol	110.11		Sigma Aldrich >99%
2-Methylphenol 95-48-7	o-Cresol	108.14		Sigma Aldrich >99%
3-Methylphenol 108-39-4	m-Cresol	108.14		Sigma Aldrich >99%
4-Methylphenol 106-44-5	p-Cresol	108.14		Sigma Aldrich >99%
4-Hydroxy-3-methoxybenzaldehyde 121-33-5	Vanillin	152.15		Sigma Aldrich >99%
Ethanol 64-17-5		46.07		Carlo Erba absolute anhydrous

2. Material and methods

2.1. Chemicals

Phenol (CAS 108-95-2), guaiacol (CAS 90-05-1), syringol (CAS 91-10-1), pyrocatechol (CAS 120-80-9), o-cresol (CAS 95-48-7), m-cresol (CAS 108-39-4), p-cresol (CAS 106-44-5) and vanillin (CAS 121-33-5) were supplied by Sigma Aldrich. All chemicals were used as received without any further purification. Deionized water and absolute ethanol were used for all experiments. Table 2 summarizes the parameters of the compounds used in this work.

2.2. Solubility of the phenolic compounds in water

Aqueous solubility of the phenolic compounds was determined using the shake-flask method between 293 and 323K. Saturated solutions of each phenolic compound in water were prepared and put in a thermostatic water bath to maintain a constant temperature in the solutions. The temperature was measured with a RTD-126U thermometer (± 0.1 K). The mixtures were stirred for 48 h with a coated magnetic stirring bar. Then, the solutions were left for another 48 h without stirring to allow the phases to settle down. Samples of the aqueous phase were taken with a syringe. In the case of solid-liquid equilibria, a 0.45- μ m filter was connected to the syringe. Samples were diluted by factor 10^4 with deionized water

before analysis. Quantification of the fraction of the phenolic compound was made using a UV-visible spectrophotometer at the maximum of absorbance for each compound. Maximum wavelengths are provided in Table S1. For each studied compounds, a calibration curve was carried out with six known concentrations solutions, comprised within 1 and 10 mg/L. Linear calibration curves (absorbance A versus the concentration C) provided R^2 correlation coefficients of 0.998 or higher.

2.3. Solubility of water in the phenolic solution

In the case of liquid-liquid equilibria between water and phenolic compound (i.e. phenol, guaiacol, o-, m-, p-cresol), the determination of the fraction of water in the phenolic phase was determined by the trouble point method [45] between 293 and 323K. An amount of phenolic compound was put into a jacketed glass cell and connected to a thermostatic bath. The temperature was measured with a RTD-126U thermometer ($\pm 0.1K$) and maintained constant all over the experiment. Solutions were stirred with a coated magnetic bar and known amounts of water were added stepwise until a second phase appears.

2.4. Vapor-liquid equilibria of phenolic compound and ethanol

Apparatus used for the VLE measurements was described in our previous work [46]. Fig. 1 presents the device used for the determination of the bubble point of isobaric VLE. A volume of 50 mL of a known mixture of phenolic compound and ethanol was introduced into a round-bottom flask (6). The flask was put into a controlled oil bath (7) and connected to a condenser (3) cooled by a glycol aqueous solution. This condenser was kept at a sufficiently low temperature (267.15 K) to minimize the amount of vapor lost through the vacuum pump section, and hence, to avoid variation of the initial composition of the sample solution. To confirm the measuring accuracy, the seal fittings in the device had to be kept in good connection to avoid air leakage. Then, the pressure was set at the desired value with the pump (11). The temperature was measured by a temperature sensor (T900 series) thermocouple calibrated with high measuring accuracy ($T900 \pm 0.06^\circ C$) and the pressure was determined by a pressure transducer (Druck-PMP4010) with an accuracy of (± 0.4 mbar). The mixture was stirred by a coated magnetic bar. After 15 min the heating has been

turned on, the temperature of the solution was generally stable and the vapor-liquid equilibrium was reached.

2.5. Excess enthalpy between phenolic compounds and ethanol

Excess molar enthalpies were measured using a Setaram C80 differential calorimeter (Setaram Instrumentation, France, 2010). The apparatus was calibrated using the Joule effect (electrical calibration) on a vessel containing a platinum resistance. Calibration was performed using several precise heat inputs to simulate experimental samples [47–49]. The phenolic compound and ethanol were initially in two compartments of the cell, separated by a mercury joint. The mixing was obtained by reversing the calorimeter. The calorimeter principle and the scheme of the cell have been described elsewhere [50].

3. Theory/calculation

3.1. Thermodynamic correlation

The phase equilibrium data of the binary systems were represented using the NRTL [44] model. This thermodynamic model allows to calculate the activity coefficient γ_i , for any component i of the binary system using the following expression:

$$\ln \gamma_i = \frac{\sum_{j=1}^m \tau_{ji} G_{ji} x_j}{\sum_{l=1}^m G_{li} x_l} + \sum_{j=1}^m \frac{x_j G_{ij}}{\sum_{l=1}^m G_{lj} x_l} \left(\tau_{ij} - \frac{\sum_{r=1}^m x_r \tau_{rj} G_{rj}}{\sum_{l=1}^m G_{lj} x_l} \right) \quad (1)$$

with $G_{ji} = \exp(-\alpha_{ji} \tau_{ji})$, $\tau_{ji} = \frac{g_{ji} - g_{ii}}{RT} = \frac{\Delta g_{ji}}{RT}$, $\Delta g_{ij} = a_{ij} + b_{ij}T$ and $\alpha_{ji} = \alpha_{ij} = \alpha$ where: g is an energy parameter characterizing the interaction of species i and j , x_i is the mole fraction of component i , T the temperature and R the ideal gas constant. The value of the non-randomness parameter α was set to 0.3 according to the literature [51]. The use of NRTL model requires the knowledge of two sets of parameters a_{ij} and b_{ij} . The parameters were determined from experimental data, and by minimizing the following objective function:

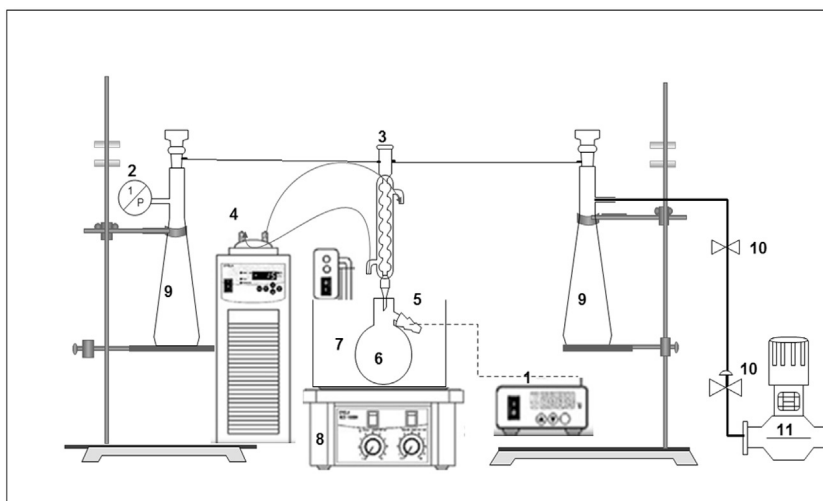


Fig. 1. Scheme of the VLE device [46]; 1. Temperature transmitter; 2. Pressure transducer (Druck-PMP4010); 3. Condenser; 4. Refrigerator; 5. Temperature sensor (T900 series); 6. Equilibrium vessel; 7. Constant temperature bath; 8. Magnetic stirrer; 9. Pressure buffer; 10. Vacuum control valve; 11 Vacuum pump.

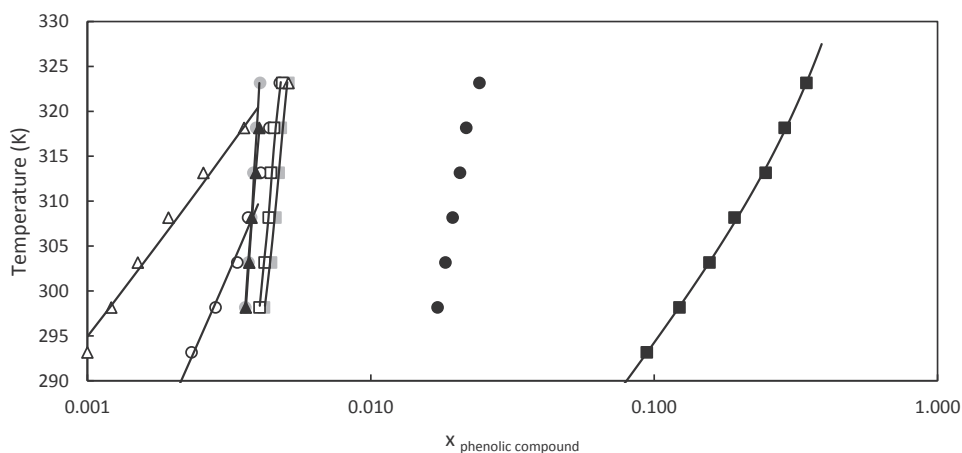


Fig. 2. Solubility of phenolic compounds in water. Experimental measurements: ● Phenol, ○ Guaiacol, ○ Syringol, ■ Pyrocatechol, □ o-Cresol, □ m-Cresol, ▲ p-Cresol, △ Vanillin. The solid lines have been calculated using NRTL model.

Table 3

Experimental LLE data of the {phenolic compound (PC) - water} systems at atmospheric pressure.

Temperature (K)	Aqueous phase		Phenolic phase	
	x _{PC}	x _{water}	x _{PC}	x _{water}
<i>Phenol-water</i>				
298.15	0.0172	0.9828		
303.15	0.0184	0.9816		
308.15	0.0195	0.9805		
313.15	0.0207	0.9793		
318.15	0.0217	0.9783		
323.15	0.0242	0.9758		
<i>Guaiacol-water</i>				
298.15	0.0036	0.9964	0.7522	0.2478
303.15	0.0037	0.9963	0.7346	0.2654
308.15	0.0038	0.9962	0.7226	0.2774
313.15	0.0039	0.9961	0.6952	0.3048
318.15	0.0039	0.9961	0.6739	0.3261
323.15	0.0041	0.9959	0.6443	0.3557
<i>Syringol-Water</i>				
313.15	0.0041	0.9959	0.3516	0.6484
318.15	0.0044	0.9956	0.3278	0.6722
323.15	0.0048	0.9952	0.3136	0.6864
<i>o-Cresol-water</i>				
298.15	0.0042	0.9958	0.5152	0.4848
303.15	0.0044	0.9956	0.5102	0.4898
308.15	0.0046	0.9954	0.5046	0.4954
313.15	0.0047	0.9953	0.4990	0.5010
318.15	0.0048	0.9952	0.4944	0.5056
323.15	0.0051	0.9949	0.4846	0.5154
<i>m-Cresol-water</i>				
298.15	0.0041	0.9959	0.5209	0.4791
303.15	0.0042	0.9958	0.5087	0.4913
308.15	0.0044	0.9956	0.4998	0.5002
313.15	0.0044	0.9956	0.4859	0.5141
318.15	0.0046	0.9954	0.4692	0.5308
323.15	0.0049	0.9951	0.4573	0.5427
<i>p-Cresol-water</i>				
298.15	0.0036	0.9964	0.4843	0.5157
303.15	0.0037	0.9963	0.4720	0.5280
308.15	0.0038	0.9962	0.4617	0.5383
313.15	0.0039	0.9961	0.4518	0.5482
318.15	0.0040	0.9960	0.4455	0.5545

u(T) = 0.1K, u(x) < 0.005.

$$F_{obj} = \sum_{k=1}^N \sum_{i=1}^2 \left\{ \left(x_{i,k}^{I,exp} - x_{i,k}^{I,calc} \right)^2 + \left(x_{i,k}^{II,exp} - x_{i,k}^{II,calc} \right)^2 \right\} \quad (2)$$

Where N is the number of tie lines in the data set, $x_{i,k}^{I,exp}$, $x_{i,k}^{I,calc}$ are the experimental and calculated mole fractions of the aqueous phase and $x_{i,k}^{II,exp}$, $x_{i,k}^{II,calc}$ are the experimental and calculated mole fractions of the phenolic phase.

The rmsd values, which provide a measure of the accuracy of the correlations, were calculated according to the following equation:

$$rmsd = \left\{ \frac{\sum_{k=1}^N \sum_{i=1}^2 \left\{ \left(x_{i,k}^{I,exp} - x_{i,k}^{I,calc} \right)^2 + \left(x_{i,k}^{II,exp} - x_{i,k}^{II,calc} \right)^2 \right\}}{4N} \right\}^{\frac{1}{2}} \quad (3)$$

Table 4

Experimental SLE data of the {phenolic compound (PC)-water} systems at atmospheric pressure.

Temperature (K)	x _{PC}	x _{water}
<i>Syringol + water</i>		
293.15	0.0023	0.9977
298.15	0.0028	0.9972
303.15	0.0034	0.9966
308.15	0.0037	0.9963
<i>Pyrocatechol + water</i>		
293.15	0.0942	0.9058
298.15	0.1229	0.8771
303.15	0.1568	0.8432
308.15	0.1923	0.8077
313.15	0.2477	0.7523
318.15	0.2892	0.7108
323.15	0.3450	0.6550
<i>Vanillin + water</i>		
293.15	0.0010	0.9990
298.15	0.0012	0.9988
303.15	0.0015	0.9985
308.15	0.0019	0.9981
313.15	0.0026	0.9974
318.15	0.0036	0.9964
323.15	0.0051	0.9949

u(T) = 0.1K, u(x) < 0.005.

3.2. Solid-liquid equilibria

The solubility of phenolic compounds in water via solid-liquid equilibria (ie pyrocatechol, syringol and vanillin) were determined using the simplified thermodynamic equation [52]:

$$\ln(x_{PC}\gamma_{PC}) = \frac{\Delta_{fus}H_{PC}}{RT_{fus,PC}} \left(1 - \frac{T_{fus,PC}}{T}\right) \quad (4)$$

where $\Delta_{fus}H_{PC}$ is the melting enthalpy calculated at the melting temperature $T_{fus,PC}$ of the phenolic compound. The objective function and the rmsd definition were similar to the equations (2) and (3). The solid phase was supposed to be pure phenolic component.

3.3. Vapor-liquid equilibria

The vapor-liquid equilibria was determined through the following expression:

$$P \cdot y_i = P_i^{sat} \cdot x_i \cdot \gamma_i \cdot \mathcal{P} \quad (5)$$

where P and P_i^{sat} are respectively the pressure and saturated pressure of compound i , with y and x the molar fractions in the vapor and liquid phases, respectively. The Poynting coefficient $\mathcal{P}$ was supposed equal to 1 since the fixed pressure was below the atmospheric one. The saturated pressures of the components were calculated using the Antoine equation [53–55] (Table S2).

The objective function used to regress the a_{ij} and b_{ij} parameter is presented as follows:

$$F_{obj} = \text{abs} \left(1 - \frac{P_{calculated}}{P_{experimental}} \right) \quad (6)$$

The corresponding rmsd was calculated with the expression:

$$\text{rmsd} = \left\{ \frac{\sum_{i=1}^N (P_{i,experimental} - P_{i,calculated})^2}{N} \right\}^{\frac{1}{2}} \quad (7)$$

4. Results and discussion

4.1. Solubility of the phenolic compounds in water

For a better display, the solubilities of the phenolic compounds in water were plotted in Fig. 2 with a natural log scale. Experimental data are provided in Tables 3 and 4. The solubility of each compounds in water increases with the increasing of temperature. Pyrocatechol, vanillin and syringol present a solid-liquid equilibria while the other compounds are in liquid-liquid equilibria with water. Note that syringol presents also a liquid-liquid equilibria but for a temperature beyond 308.15K. The solubility of pyrocatechol, phenol and vanillin is the most influenced by temperature. At 298.15 K, pyrocatechol presents the highest solubility in water ($x_{\text{pyrocatechol}} = 0.123$), followed by phenol ($x_{\text{phenol}} = 0.017$). The solubility of the others compounds are in the same range of values, according to the following rank: o-cresol > m-cresol > p-cresol > guaiacol > syringol > vanillin. The pyrocatechol and phenol molecules present respectively two and one hydroxyl groups, which are easily accessible to interact with water. In the case of cresols, the presence of a methyl group hinders the interactions of water with the hydroxyl group and therefore reduces the solubility of the compound. Finally, for guaiacol, syringol and vanillin, the presence of a methoxy group creates an intramolecular hydrogen bond with the hydroxyl group of the phenolic compounds. For these three former molecules, their hydroxyl groups are less available to interact with water. Generally, measurements are in good agreement with those found in literature [20–34].

Table 5
NRTL parameters for the phenolic compound –water (SLE and LLE) systems.

Compounds	a_{12} (J.mol ⁻¹)	a_{21} (J.mol ⁻¹)	b_{12} (J.mol ⁻¹ .K ⁻¹)	b_{21} (J.mol ⁻¹ .K ⁻¹)	rmsd (%)
Guaiacol	–2904.47	14631.14	50.56	–41.82	0.1363
o-Cresol	2238.76	4062.37	36.27	–14.81	0.0403
m-Cresol	–678.03	8057.33	46.23	–27.99	0.0298
p-Cresol	–839.66	6804.87	48.39	–25.14	0.0399
Syringol ^a	–4.48	–4.73	40.04	92.25	0.0041
Pyrocatechol ^a	2049.51	18565.35	182.94	–60.35	0.1488
Vanillin ^a	23464.72	5611.93	–30.50	2754755.83	0.0085

^a Solid-liquid equilibrium.

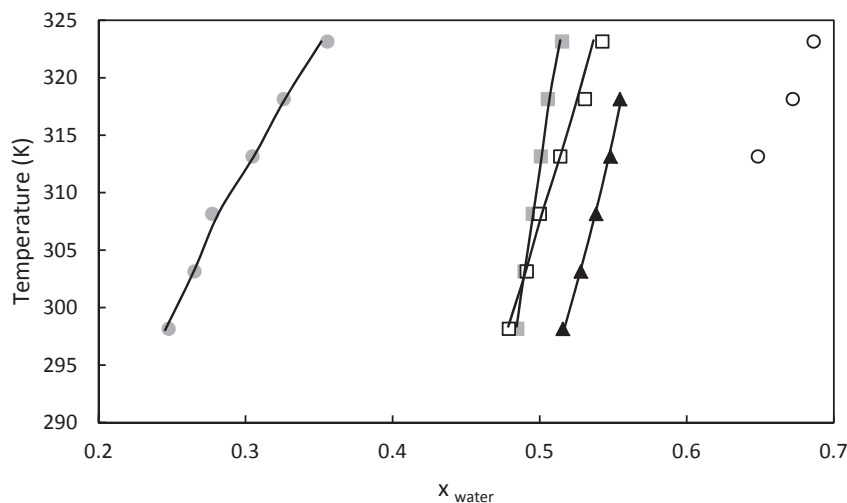


Fig. 3. Water solubility in phenolic phases. Experimental measurements: ● Guaiacol, ○ Syringol, ■ o-Cresol, □ m-Cresol, ▲ p-Cresol. The solid lines have been calculated using NRTL model.

Table 6
Vapor-liquid equilibrium data of binary mixtures containing phenolic compounds and ethanol.

P (bar)	x_{phenol}	T (K)	P (bar)	x_{guaiacol}	T (K)	P (bar)	$x_{\text{o-cresol}}$	T (K)
0.090	0.100	303.09	0.090	0.109	300.25	0.090	0.097	302.23
	0.198	305.10		0.219	303.33		0.206	305.46
	0.300	310.04		0.322	304.88		0.304	308.67
	0.400	313.16		0.400	307.11		0.390	312.21
	0.495	319.73		0.491	309.40		0.507	319.01
	0.588	325.77		0.592	312.76		0.597	326.79
	0.692	333.65		0.701	316.55		0.699	333.07
	0.794	342.26		0.773	320.29		0.791	342.80
	0.883	353.85		0.889	332.80		0.898	359.99
0.150	0.100	312.58	0.164	0.109	313.11	0.150	0.097	311.87
	0.198	315.16		0.219	315.03		0.206	315.32
	0.300	319.57		0.322	317.28		0.304	319.29
	0.400	323.66		0.400	319.36		0.390	322.19
	0.495	330.26		0.491	321.93		0.507	329.86
	0.588	337.31		0.592	325.08		0.597	336.50
	0.692	345.08		0.701	329.79		0.699	344.12
	0.794	355.05		0.773	333.56		0.791	354.31
	0.883	366.57		0.889	347.75		0.898	373.10
0.330	0.100	328.70	0.333	0.109	328.13	0.351	0.097	329.36
	0.198	331.35		0.219	329.87		0.206	333.25
	0.300	336.15		0.322	332.29		0.304	337.22
	0.400	339.90		0.400	334.17		0.390	340.16
	0.495	347.78		0.491	337.32		0.507	348.90
	0.588	355.56		0.592	340.83		0.597	356.03
	0.692	364.32		0.701	346.20		0.699	363.84
	0.794	376.11		0.773	350.90		0.791	375.90
	0.883	388.50		0.889	368.35	—	—	—
0.550	0.100	340.09	0.500	0.109	337.36	0.551	0.097	339.54
	0.198	342.95		0.219	339.06		0.206	343.48
	0.300	348.05		0.322	341.49		0.304	347.87
	0.400	352.38		0.400	343.46		0.390	350.35
	0.495	360.94		0.491	346.85		0.507	360.29
	0.588	369.17		0.592	350.46		0.597	367.82
	0.692	378.58		0.701	356.72		0.699	376.33
	0.794	389.90		0.773	362.19		0.791	389.39
—	—	—	—	0.889	381.56	—	—	—
0.750	0.100	347.53	0.700	0.109	345.35	0.750	0.097	346.85
	0.198	350.61		0.219	347.10		0.206	351.09
	0.300	355.78		0.322	349.65		0.304	355.53
	0.400	360.18		0.400	351.60		0.390	357.50
	0.495	369.28		0.491	355.68		0.507	368.40
	0.588	378.05		0.592	359.53		0.597	376.10
	0.692	387.67		0.701	366.51		0.699	385.84
—	—	—	—	0.773	372.24	—	—	—
—	—	—	—	0.889	394.05	—	—	—
0.851	0.100	350.58	0.832	0.109	349.63	0.851	0.097	349.95
	0.198	353.70		0.219	351.45		0.206	354.17
	0.300	359.06		0.322	354.00		0.304	358.80
	0.400	363.56		0.400	355.80		0.390	360.75
	0.495	373.06		0.491	360.20		0.507	371.88
	0.588	381.78		0.592	364.10		0.597	379.52
—	—	—	—	0.701	371.89	—	0.699	389.62
—	—	—	—	0.773	377.68	—	—	—
—	—	—	—	0.889	400.47	—	—	—
0.986	0.100	354.51	0.996	0.109	354.12	0.991	0.097	353.75
	0.198	357.65	0.961	0.219	354.92	0.977	0.206	357.95
	0.300	362.97	0.992	0.322	358.70	0.997	0.304	363.05
	0.400	367.75	0.977	0.400	359.87	0.999	0.390	364.60
	0.495	376.85	0.990	0.491	365.43	0.991	0.507	374.97
	0.588	386.30	0.971	0.592	368.80	0.985	0.597	384.56
—	—	—	0.997	0.701	377.89	0.994	0.699	394.11
—	—	—	0.990	0.773	383.82	—	—	—
—	—	—	0.965	0.889	409.48	—	—	—

$u(T) = 0.06$ K, $u(P) = 0.4$ mbar, $u(x) < 0.005$.

Nevertheless, our experimental data obtained for the vanillin-water system presents some deviation with those measured by Noubigh et al. [28]. Indeed, this difference can be explained by the fact these authors waited only 3 h of stirring and did not reach the equilibria [57–59]. Besides, for the vanillin-water system, the results presented in this work are in good agreement with those from Cartwright [26] and Mange et al. [27]. Nevertheless, our experimental data for the guaiacol-water system is in disagreement with those found in the work of Jaspersion et al. [56]. For the solubility of syringol in water, no data have been found in the literature. Plot of the different data are provided in supporting information for comparison (Figs. S1–S7).

4.2. Solubility of water in the phenolic solution

Tables 3 and 4 and Fig. 3 gather the results of the solubility of water in the phenolic solutions. Water is more soluble in the phenolic solutions with the rise of temperature. The guaiacol system is the most temperature-dependent. At 300 K, the solubility of water increases in the phenolic compounds as follows: guaiacol < o-cresol < m-cresol < p-cresol < syringol. The solubility of water in guaiacol is very low compared to those obtained in cresols or in syringol. This behavior is mainly due to the fact that guaiacol presents in its structure an intramolecular hydrogen bond between its hydroxyl and methoxy group [60]. Therefore, the hydroxyl group is less accessible to interact with water. In the case of cresols, the solubility of water increases with the increasing distance of the methyl group. The hydroxyl group is less hindered in the para position compared to the ortho position and therefore can interact with water more easily. In the case of syringol, there is an intramolecular hydrogen bond between its hydroxyl and methoxy groups but its second methoxy group remains available to interact with water molecules. Experimental solubility measurements are in good agreements with those found in the literature [21,23–25,35] with a deviation between 0.15 and 1.4%. Comparisons are provided in supporting information (Figs. S4–S7).

The regression of the parameter a_{ij} and b_{ij} of the NRTL model are listed in Table 5 for the phenolic compound-water systems. It can be noted that the number of experimental data of the two binary systems {phenol-water} and {syringol-water} does not allow to regress the NRTL parameters. The rmsd values are very low (inferior to 0.3%) for all systems which means that the correlated values fit well with the experimental data (Figs. 2 and 3).

4.3. Vapor-liquid equilibria of phenolic compound and ethanol

The results of the vapor-liquid equilibria between the phenolic compounds (*i.e.* phenol, guaiacol, o-cresol) and ethanol are summarized hereafter (Table 6). The boiling temperature increases with the increase of pressure. Same phenomenon is observed with the increase of the molar fraction of the phenolic compounds. No azeotrope point is observed for the three systems, neither is a demixing of the liquid phase. Fig. 4 presents the plot of the guaiacol-ethanol system as example.

The temperature dependency is more important at high fraction of phenolic compounds ($x_{\text{phenolic}} = 0.8$ –1). The binary system {guaiacol-ethanol} is less temperature-dependent compared to the phenol or o-cresol systems (Figs. S8 and S9). Indeed, the slope in the guaiacol system is higher compared to the other systems. Moreover, for the {o-cresol-ethanol} and {phenol-ethanol} systems, the increase of the boiling temperature is more regularly distributed on the range of the molar fraction studied. The final boiling

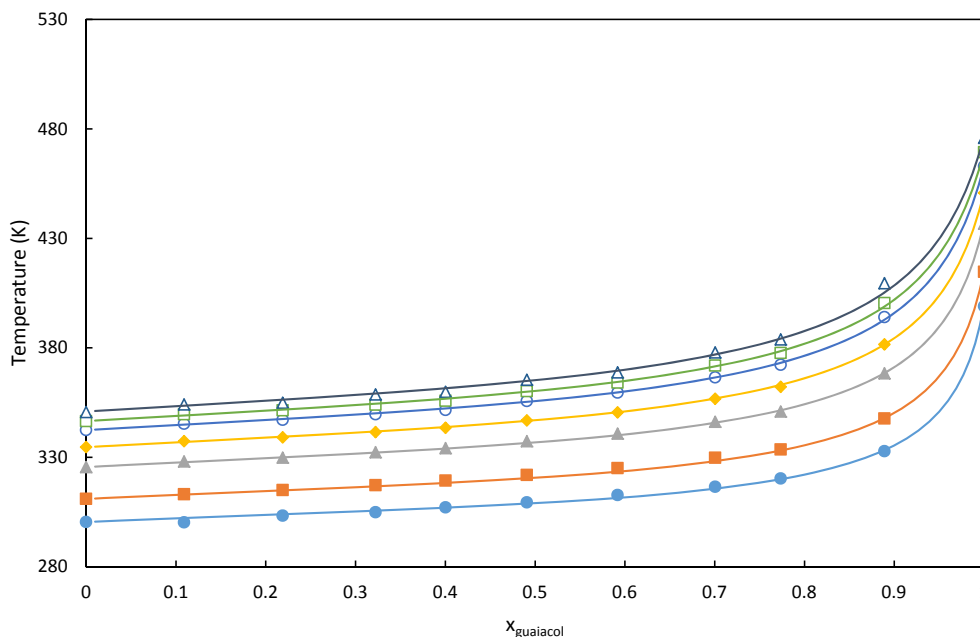


Fig. 4. Vapor-Liquid equilibria of the guaiacol-ethanol system in function of the molar fraction of guaiacol at different pressures. Experimental measurements at pressures: ● 0.09 bar, ■ 0.165 bar, ▲ 0.335 bar, ◆ 0.5 bar, ○ 0.7 bar, □ 0.83 bar, △ 0.99 bar. The solid lines have been calculated using NRTL model.

Table 7
NRTL parameters for the phenolic compound – ethanol (VLE) systems.

Compounds	a_{12} (J.mol ⁻¹)	a_{21} (J.mol ⁻¹)	b_{12} (J.mol ⁻¹ .K ⁻¹)	b_{21} (J.mol ⁻¹ .K ⁻¹)	rmsd (bar)
Phenol	−0.04	−10.99	0.01	5.47	0.021
Guaiacol	3327.95	−9.21	2215.75	−1.38	0.022
o-Cresol	142951.43	−4.65	−4284.01	5.11	0.025

temperature is decreasing depending on the phenolic compounds as follows: guaiacol > o-cresol > phenol.

The regression of the parameter a_{ij} and b_{ij} of the NRTL model are listed in Table 7 for the phenolic compound-ethanol system. The rmsd values are very low (inferior to 0.02 bar) for all systems showing that the correlated values fit well with the experimental data.

4.4. Excess enthalpy of the phenolic compound – ethanol systems

Excess properties give information for understanding the origin of the nonideality of mixtures. Molar excess enthalpy (h^E) provides a measure of the intermolecular interactions in the mixture. During mixing, energy is required to break the interactions between the molecules of the pure solvent in order to form new interactions between unlike species releasing energy. The results of the molar excess enthalpy (h^E) for guaiacol, o-cresol and m-cresol and ethanol systems are listed in Table 8 and plotted in Fig. 5. The mixing of ethanol and phenolic compounds studied in this work is exothermic and the curves are practically symmetrical towards the molar fraction $x = 0.5$. In first approximation, the exothermic effect may be related to stronger interactions between phenolic compounds and ethanol than ethanol-ethanol or phenolic-phenolic compounds interactions. Finally, the mixing enthalpy values

Table 8
Molar excess enthalpy of the {phenolic compound – ethanol} systems at $T = 298.15$ K and $P = 1$ atm.

x_{Guaiacol}	h^E (J.mol ⁻¹)	$x_{\text{o-cresol}}$	h^E (J.mol ⁻¹)	$x_{\text{m-cresol}}$	h^E (J.mol ⁻¹)
0.0168	−29.95	0.1954	−1112.03	0.2003	−791.99
0.1988	−379.49	0.4127	−1620.73	0.4097	−1227.70
0.3994	−549.10	0.6007	−1652.17	0.5861	−1146.24
0.6080	−505.33	0.8013	−943.58	0.7875	−687.42
0.7783	−237.77	—	—	—	—

$u(T) = 0.02$ K; $u(H^E) = 5$ J.mol⁻¹; $u(x) < 0.005$.

decreases as follows: guaiacol > m-cresol > o-cresol. It is worth noting that molar excess enthalpy of {ethanol + guaiacol} is quite small (0–550 J mol⁻¹). Replacing the methoxy group of the guaiacol by a methyl group in o-cresol leads to a strong increase of the h^E values.

5. Conclusion

The mutual solubilities of water and eight phenolic compounds (i.e. phenol, guaiacol, syringol, pyrocatechol, o-, m-, p-cresol and vanillin) were investigated between 293.15 and 323.15 K. Vapor-liquid equilibria and mixing enthalpy of binary mixtures {phenol or guaiacol or o-cresol + ethanol} were also investigated for different molar compositions. The information provided by this work help to understand the behavior of the phenolic compounds in green solvents. This knowledge is vital for the development of extraction and purification processes of phenolic compounds from bio-oils produced upon biomass thermochemical conversion. Moreover, parameters of the NRTL thermodynamic model were determined and correlate well with the experimental measurements.

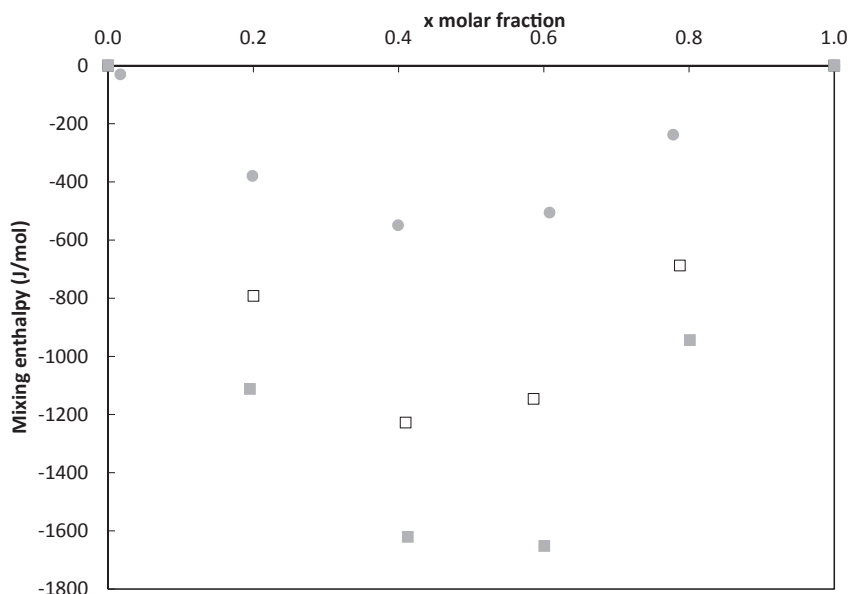


Fig. 5. Mixing enthalpy of the phenolic compound - ethanol systems. ● Guaiacol, ■ o-Cresol, □ m-Cresol.

Acknowledgment

The authors would like to acknowledge French government for the Grant for PhD thesis. This research did not receive any other specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

List of symbols

α	Non-randomness parameter
γ_i	Activity coefficient
a_{ij}, b_{ij}	NRTL parameters
g	Interaction energy parameter (NRTL)
h^E	Molar excess enthalpy
$\Delta_{fus}H$	Fusion enthalpy
K	Kelvin
LLE	Liquid-Liquid Equilibrium
N	Number of tie-lines
P	Poynting coefficient
P	Pressure
p^{sat}	Saturated pressure
PC	Phenolic Compound
R	Ideal gas constant
rmsd	Root mean square deviation
SLE	Solid Liquid Equilibrium
T	Temperature
T_{fus}	Fusion temperature
VLE	Vapor-Liquid Equilibrium
x	Molar fractions (liquid phase)
y	Molar fraction (vapor phase)

Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.fluid.2017.09.008>.

References

- [1] A.V. Bridgwater, D. Meier, D. Radlein, An overview of fast pyrolysis of biomass, *Org. Geochem.* 30 (1999) 1479–1493, [http://dx.doi.org/10.1016/S0146-6380\(99\)00120-5](http://dx.doi.org/10.1016/S0146-6380(99)00120-5).
- [2] R.K. Sharma, N.N. Bakhshi, Catalytic upgrading of biomass-derived oils to transportation fuels and chemicals, *Can. J. Chem. Eng.* 69 (1991) 1071–1081, <http://dx.doi.org/10.1002/cjce.5450690505>.
- [3] M.D. Guillén, M.L. Ibargoitia, New components with potential antioxidant and organoleptic properties, detected for the first time in liquid smoke flavoring preparations, *J. Agric. Food Chem.* 46 (1998) 1276–1285, <http://dx.doi.org/10.1021/jf970952x>.
- [4] F. Helmut, V. Heinz-Werner, H. Toshikazu, P. Wilfried, Phenol derivatives, *Ullmann's encycl. Ind. Chem. VCH Ger.* 19 (1985) 299–357.
- [5] F. Ikegami, T. Sekine, Y. Fujii, Anti-dermatophyte activity of phenolic compounds in "mokusaku-eki", *Yakugaku Zasshi* 118 (1998) 27–30.
- [6] J.A. Maga, I. Katz, Simple phenol and phenolic compounds in food flavor, *C.R. Crit. Rev. Food Sci. Nutr.* 10 (1978) 323–372, <http://dx.doi.org/10.1080/10408397809527255>.
- [7] X. Ouyang, X. Huang, Y. Zhu, X. Qiu, Ethanol-enhanced liquefaction of lignin with formic acid as an *in situ* hydrogen donor, *Energy Fuels* 29 (2015) 5835–5840, <http://dx.doi.org/10.1021/acs.energyfuels.5b01127>.
- [8] T. Wahyudiono, M. Kanetake, M. Sasaki, Goto, decomposition of a lignin model compound under hydrothermal conditions, *Chem. Eng. Technol.* 30 (2007) 1113–1122, <http://dx.doi.org/10.1002/ceat.200700066>.
- [9] M. Wahyudiono, M. Sasaki, Goto, Conversion of biomass model compound under hydrothermal conditions using batch reactor, *Fuel* 88 (2009) 1656–1664, <http://dx.doi.org/10.1016/j.fuel.2009.02.028>.
- [10] X. Yuan, H. Li, G. Zeng, J. Tong, W. Xie, Sub- and supercritical liquefaction of rice straw in the presence of ethanol–water and 2-propanol–water mixture, *Energy* 32 (2007) 2081–2088, <http://dx.doi.org/10.1016/j.energy.2007.04.011>.
- [11] H.-M. Liu, Y.-L. Liu, Characterization of milled solid residue from cypress liquefaction in sub- and super ethanol, *Bioresour. Technol.* 151 (2014) 424–427, <http://dx.doi.org/10.1016/j.biortech.2013.10.050>.
- [12] R. Singh, A. Prakash, S.K. Dhiman, B. Balagurumurthy, A.K. Arora, S.K. Puri, T. Bhaskar, Hydrothermal conversion of lignin to substituted phenols and aromatic ethers, *Bioresour. Technol.* 165 (2014) 319–322, <http://dx.doi.org/10.1016/j.biortech.2014.02.076>.
- [13] C. Xu, T. Etcheverry, Hydro-liquefaction of woody biomass in sub- and supercritical ethanol with iron-based catalysts, *Fuel* 87 (2008) 335–345, <http://dx.doi.org/10.1016/j.fuel.2007.05.013>.
- [14] Z. Liu, F.-S. Zhang, Effects of various solvents on the liquefaction of biomass to produce fuels and chemical feedstocks, *Energy Convers. Manag.* 49 (2008) 3498–3504, <http://dx.doi.org/10.1016/j.enconman.2008.08.009>.
- [15] J.N. Murwanashyaka, H. Pakdel, C. Roy, Separation of syringol from birch wood-derived vacuum pyrolysis oil, *Sep. Purif. Technol.* 24 (2001) 155–165, [http://dx.doi.org/10.1016/S1383-5866\(00\)00225-2](http://dx.doi.org/10.1016/S1383-5866(00)00225-2).
- [16] C. Amen-Chen, H. Pakdel, C. Roy, Separation of phenols from Eucalyptus wood tar, *Biomass Bioenergy* 13 (1997) 25–37, [http://dx.doi.org/10.1016/S0961-9534\(97\)00021-4](http://dx.doi.org/10.1016/S0961-9534(97)00021-4).
- [17] R.W. Thring, J. Breau, Hydrocracking of solvolysis lignin in a batch reactor, *Fuel* 75 (1996) 795–800, [http://dx.doi.org/10.1016/0016-2361\(96\)00036-1](http://dx.doi.org/10.1016/0016-2361(96)00036-1).
- [18] S. Wang, Y. Wang, Q. Cai, X. Wang, H. Jin, Z. Luo, Multi-step separation of monophenols and pyrolytic lignins from the water-insoluble phase of bio-oil, *Sep. Purif. Technol.* 122 (2014) 248–255, <http://dx.doi.org/10.1016/j>

- j.seppur.2013.11.017.
- [19] S. Czernik, R. Maggi, G.V.C. Peacocke, A review of physical and chemical methods of upgrading biomass-derived fast pyrolysis liquids, in: *Biomass Proc. 4th Biomass Conf. Am.*, 1999, pp. 1235–1240.
 - [20] D. Tam, D. Varhanickova, W.Y. Shiu, D. Mackay, Aqueous solubility of chloroguaiacols, *J. Chem. Eng. Data* 39 (1994) 83–86, <http://dx.doi.org/10.1021/je00013a022>.
 - [21] N.V. Sidgwick, W.J. Spurrell, T.E. Davies, CXXXII.—the solubility of the nitrophenols and other isomeric disubstitution products of benzene, *J. Chem. Soc. Trans.* 107 (1915) 1202–1213, <http://dx.doi.org/10.1039/CT9150701202>.
 - [22] D. Varhanickova, W.-Y. Shiu, D. Mackay, Aqueous solubilities of alkylphenols and methoxyphenols at 25.degree.C, *J. Chem. Eng. Data* 40 (1995) 448–451, <http://dx.doi.org/10.1021/je00018a020>.
 - [23] C.R. Bailey, CCXCVIII.—the increased solubility of phenolic substances in water on addition of a third substance, *J. Chem. Soc. Trans.* 123 (1923) 2579–2590, <http://dx.doi.org/10.1039/CT9232302579>.
 - [24] H. Brusset, H. Gillier-Pandraud, A. Neuman, Strukturbestimmung bei-150 grad c von 2, 5- und 2, 3-dimethylphenol, *Chem. Inf.* 3 (1972).
 - [25] W.A. Leet, H.-M. Lin, K.-C. Chao, Mutual solubilities in six binary mixtures of water + a heavy hydrocarbon or a derivative, *J. Chem. Eng. Data* 32 (1987) 37–40, <http://dx.doi.org/10.1021/je00047a010>.
 - [26] L.C. Cartwright, Vanilla-like synthetics, solubility and volatility of propenyl guaiethyl, bourbonal, vanillin, and coumarin, *J. Agric. Food Chem.* 1 (1953) 312–314, <http://dx.doi.org/10.1021/jf60004a006>.
 - [27] C.E. Mange, O. Ehler, Solubilities of vanillin, *Ind. Eng. Chem.* 16 (1924) 1258–1260, <http://dx.doi.org/10.1021/jie50180a017>.
 - [28] A. Noubigh, M. Abderrabba, E. Provost, Temperature and salt addition effects on the solubility behaviour of some phenolic compounds in water, *J. Chem. Thermodyn.* 39 (2007) 297–303, <http://dx.doi.org/10.1016/j.jct.2006.06.014>.
 - [29] W.H. Walker, A.R. Collett, C.L. Lazzell, The solubility relations of the isomeric dihydroxybenzenes, *J. Phys. Chem.* 35 (1930) 3259–3271, <http://dx.doi.org/10.1021/j150329a011>.
 - [30] X. Xia, D. Jiang, Determination and correlation of solubilities of catechol, *J. Chem. Ind. Eng.-CHINA-* 58 (2007) 1082.
 - [31] C. Achard, M. Jaoui, M. Schwing, M. Rogalski, Aqueous solubilities of phenol derivatives by conductivity measurements, *J. Chem. Eng. Data* 41 (1996) 504–507, <http://dx.doi.org/10.1021/je950202o>.
 - [32] A.N. Campbell, A.J.R. Campbell, Concentrations, total and partial vapor pressures, surface tensions and viscosities, in the systems phenol–water and phenol–water–4% succinic acid, *J. Am. Chem. Soc.* 59 (1937) 2481–2488, <http://dx.doi.org/10.1021/ja01291a001>.
 - [33] A.E. Hill, W.M. Malisoff, The mutual solubility of liquids. III. The mutual solubility of phenol and water. IV. The mutual solubility of normal butylalcohol and water, *J. Am. Chem. Soc.* 48 (1926) 918–927, <http://dx.doi.org/10.1021/ja01415a011>.
 - [34] M. Jaoui, C. Achard, M. Rogalski, Solubility as a function of temperature of selected chlorophenols and nitrophenols in aqueous solutions containing electrolytes or surfactants, *J. Chem. Eng. Data* 47 (2002) 297–303, <http://dx.doi.org/10.1021/je0102309>.
 - [35] T. Shedlovsky, On guaiacol solutions: the electrical conductivity of sodium and potassium guaiacolate in guaiacol, *J. Gen. Physiol.* 17 (1934) 549–561, <http://dx.doi.org/10.1085/jgp.17.4.549>.
 - [36] M. Klauck, A. Grenner, K. Taubert, A. Martin, R. Meinhardt, J. Schmelzer, Vapor–Liquid equilibria in binary systems of phenol or cresols + water, + toluene, and + octane and Liquid–Liquid equilibria in binary systems of cresols + water, *Ind. Eng. Chem. Res.* 47 (2008) 5119–5126, <http://dx.doi.org/10.1021/ie071214t>.
 - [37] T.E.V. Prasad, A. Jaiswal, S. Prasad, G. Harish, N. Krupavaram, N.M. Sirisha, K. Ashok, Y.N. Kumar, N. Venkanna, D.H.L. Prasad, Activity coefficients of the binary mixtures of o-cresol or p-cresol with C1–C4 aliphatic alcohols near ambient pressure, *Fluid Phase Equilibria* 244 (2006) 86–98, <http://dx.doi.org/10.1016/j.fluid.2006.03.018>.
 - [38] G. Weissenberger, R. Henke, L. Bregmann, Zur Kenntnis organischer Molekülverbindungen: XVI. Zweiwertige Phenole und ihre Äther, *Monatsh. Chem.* 46 (1925) 471–482, <http://dx.doi.org/10.1007/BF01525702>.
 - [39] G. Weissenberger, L. Piatti, The molecular compounds of the phenols. I. The behavior of cresols against alcohol, *Monatsh. Chem.* 45 (n.d.) 187–206, doi: 10.1007/BF01524660.
 - [40] C.H. Chou, J.L. Perng, M.D. Lee, Y.P. Chen, Vapor-liquid equilibrium measurements of the ternary system of ethanol, water and phenol at 760 mm Hg, *J. Chin. Inst. Chem. En.* 18 (1987) 393–399.
 - [41] N.V. Novikova, V.N. Ponomarev, V.S. Timofeev, Investigation of the separation technology of the mixture ethanol - water - butyl acetate - phenol, *Osn. Org. Sint. Neft.* 25 (1989) 124–129.
 - [42] G. Weissenberger, F. Schuster, K. Schuler, The molecular compounds of phenols. IV. The behavior of the binary systems with phenol and phenol ethers, *Monatsh. Chem.* 45 (1924) 425–435.
 - [43] G. Weissenberger, R. Henke, E. Sperling, Zur Kenntnis organischer Molekülverbindungen XVII. Das Verhalten des Dekahydronaphthalins, *Monatsh. Chem.* 46 (1925) 483–497, <http://dx.doi.org/10.1007/BF01526344>.
 - [44] H. Renon, J.M. Prausnitz, Local compositions in thermodynamic excess functions for liquid mixtures, *AIChE J.* 14 (1968) 135–144, <http://dx.doi.org/10.1002/aic.690140124>.
 - [45] M. Rogošić, M. Bakula, M. Župan, Liquid-liquid equilibria in the ternary systems H2O–phenol–2-butanone and H2O–phenol–2-propanol, *Chem. Biochem. Eng. Q.* 26 (2012) 155–162.
 - [46] E.-S. Abumandour, F. Mutelet, D. Alonso, Performance of an absorption heat transformer using new working binary systems composed of {ionic liquid and water}, *Appl. Therm. Eng.* 94 (2016) 579–589, <http://dx.doi.org/10.1016/j.applthermaleng.2015.10.107>.
 - [47] C80, The Brochure, Setaram Instrumentation, Caluire, France, 2009.
 - [48] C80, Commissioning, Setaram Instrumentation, Caluire, France, 2010.
 - [49] I. Letyanina, N. Tsvetov, I. Zvereva, A. Samarov, A. Toikka, Excess molar enthalpies for binary mixtures of n-propanol, acetic acid, and n-propyl acetate at 313.15K and atmospheric pressure, *Fluid Phase Equilibria* 381 (2014) 77–82, <http://dx.doi.org/10.1016/j.fluid.2014.08.022>.
 - [50] P.M. Ghogomu, M. Bouroukba, J. Dellacherie, D. Balesdent, M. Dirand, On the ideality of liquid mixtures of long-chain n-alkanes, *Thermochim. Acta* 306 (1997) 69–71, [http://dx.doi.org/10.1016/S0040-6031\(97\)00301-8](http://dx.doi.org/10.1016/S0040-6031(97)00301-8).
 - [51] L.D. Simoni, A. Chapeaux, J.F. Brennecke, M.A. Stadtherr, Asymmetric framework for predicting Liquid–Liquid equilibrium of ionic liquid–mixed-solvent systems. 2. Prediction of ternary systems, *Ind. Eng. Chem. Res.* 48 (2009) 7257–7265, <http://dx.doi.org/10.1021/ie9004628>.
 - [52] J.M. Prausnitz, R.N. Lichtenthaler, E.G. de Azevedo, *Molecular Thermodynamics of Fluid-phase Equilibria*, Prentice-Hall, 1986.
 - [53] <http://www.engr.umd.edu/~nsw/ench250/antoine.dat> - Antoine_coefficient_table.PDF, (n.d.). http://www.eng.auburn.edu/~drmls/mans486/Diffusion%20Tube/Antoine_coefficient_table.PDF (accessed April 12, 2017).
 - [54] E. Von Terres, F. Gebert, H. Hulsemann, P. Heinz, H. Toepsch, W. Ruppert, Zur Kenntnis der Physikalisch-Chemischen Grundlagen der Gewinnung und Zerlegung derPheolfractionen von Steibohlenteer und Braunkohlenschwelter, *Brennst. Chem.* 36 (1955) 272.
 - [55] S. Henke, P. Kadlec, Z. Bubnik, Physico-chemical properties of ethanol – compilation of existing data, *J. Food Eng.* 99 (2010) 497–504, <http://dx.doi.org/10.1016/j.jfoodeng.2009.06.050>.
 - [56] L.V. Jaspersen, R.J. McDougal, V. Diky, E. Paulechka, R.D. Chirico, K. Kroenlein, K. Iisa, A. Dutta, Liquid–liquid equilibrium measurements for model systems related to catalytic fast pyrolysis of biomass, *J. Chem. Eng. Data* 62 (2017) 243–252, <http://dx.doi.org/10.1021/acs.jced.6b00625>.
 - [57] F.L. Mota, A.J. Queimada, S.P. Pinho, E.A. Macedo, Aqueous solubility of some natural phenolic compounds, *Ind. Eng. Chem. Res.* 47 (2008) 5182–5189, <http://dx.doi.org/10.1021/ie071452o>.
 - [58] A.J. Queimada, F.L. Mota, S.P. Pinho, E.A. Macedo, Solubilities of biologically active phenolic compounds: measurements and modeling, *J. Phys. Chem. B* 113 (2009) 3469–3476, <http://dx.doi.org/10.1021/jp808683y>.
 - [59] L.-L. Lu, X.-Y. Lu, Solubilities of gallic acid and its esters in water, *J. Chem. Eng. Data* 52 (2007) 37–39, <http://dx.doi.org/10.1021/je0601661>.
 - [60] C. Agache, V.I. Popa, Ab initio studies on the molecular conformation of lignin model compounds I. Conformational preferences of the phenolic hydroxyl and methoxy groups in guaiacol, *Monatsh. Für Chem. Chem. Mon.* 137 (2006) 55–68, <http://dx.doi.org/10.1007/s00706-005-0404-x>.